INDIAN INSTITUTE OF INFORMATION TECHNOLOGY RAICHUR

Mathematics Club

PROBLEM NUMBER: WP2611

DATE: MAY 21, 2026

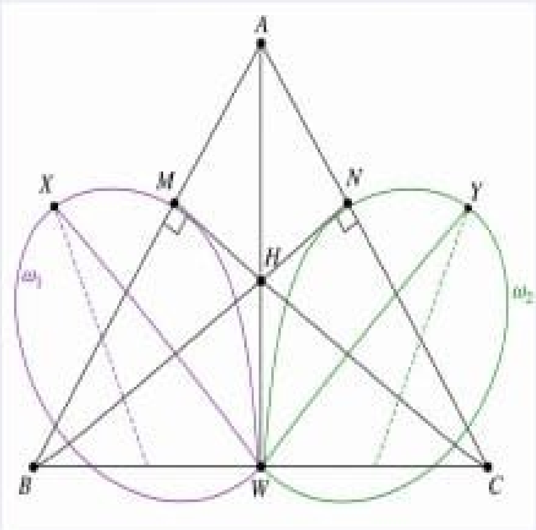
DUE DATE: MAY 31, 2026

WEEKLY PROBLEM

Circular Dilemma

EXERCISE.

Let ABC be an acute triangle with orthocenter H, and let W be a point on the side $\overline{BC}$, between B and C. The points M and N are the feet of the altitudes drawn from B and C, respectively. ω_1 is the circumcircle of the triangle BWN and X is a point such that $\overline{WX}$ is a diameter of ω_1. Similarly, ω_2 is the circumcircle of triangle CWM and Y is a point such that $\overline{WY}$ is a diameter of ω_2. Show that the points X, Y, and H are collinear.



1. **Part 1** Show that: $\angle BNW = \angle BMW = 90^\circ$ and hence prove that B, W, M, N are concyclic.
2. **Part 2**

   Since WX and WY are diameters of ω_1 and ω_2:

   (a) Prove that $\angle WNX = 90^\circ$ and $\angle WMY = 90^\circ$

   (b) Deduce that $XN||AC$ and $YM||AB$
3. **Part 3**

   Using the realtions from Part 2 and the fact that H is the orthocenter of triangle ABC, prove that the points X, Y, H are collinear.

Top Submissions for Problem (WP2610):

1. **Ruthala Jaswanth Kumar** - MC24B1036
2. **Kumar Raushan Mehta** - AD25B1015
3. **Adelly Sai Teja** - CS25B1003
4. **Yuvraj Satvik** - MC24B1004